Royal Canadian Air Cadets
110 Black Hawk Squadron
Proficiency Level 2 Aviation Exam Version B
March 25, 2024
Duration: 60 minutes

Rank: ____

Surname: ____

Given Name: _______________________________________

CIN: ____

Exam Information & Instructions: Please read carefully

This test is based on multiple-choice questions. There are **27** questions. Ensure you have all **3** pages attached. All questions carry equal weight. On the front of the Exam Sheet, ensure that you write your rank, last name, first name and cadet information number. On the Exam Sheet, you can use either pencil or pen to circle the most correct answer.

Ensure you **DO NOT** have any electronic devices (phone, smartwatch, headphones) on your person and are silenced and in your bag. Any offense will lead to a deduction of marks. Only items allowed on your desk are your pen/pencil, eraser, transparent water bottle or pencil case. Everything else (opaque containers, wedge/beret) are to be put in your bags.

Only circle ONE answer for each question, no marks will be given for questions with multiple answers selected. Cleanly erase any answer you wish to change to ensure there is only ONE clear answer. Circle in each case the best answer out of the alternatives given. You should attempt all questions.

Good Luck!

1. An older aircraft uses a rotary piston engine. What critical operational limitation would this engine impose compared to modern radial engines?
    A. Poor reliability due to constant heat generation
    B. Excessive fuel consumption at high airspeeds
    C. Inability to operate at high altitudes
    D. Heavy weight of spinning cylinders
2. During a turn, a pilot simultaneously applies elevator and aileron inputs. Which axes and are primarily involved in this maneuver?
    A. Longitudinal axis and vertical axis
    B. Vertical axis and lateral axis
    C. Lateral axis and Longitudinal axis
3. An aircraft's empennage is modified with vertical stabilizers. How does this affect its stability and control?
    A. Unwanted yaw movement intensifies
    B. Longitudinal stability weakens
    C. Drag increases
    D. None of the above
4. An aircraft design team is tasked with improving longitudinal axis stability during flight. Which control surface should they prioritize optimizing?
    A. Elevators
    B. Ailerons
    C. Rudders
    D. Flaps
5. During final approach, a pilot extends the flaps to prepare for landing. Which of the following describes the primary aerodynamic trade-off caused by this action?
    A. Increased drag and increased lift
    B. Increased lift and reduced drag
    C. Reduced lift and increased thrust
    D. None of the above
6. How many aircraft control surfaces are CORRECTLY matched to their functions?

    I. Rudder: Controls yaw around the vertical axis.

    II. Elevator: Adjusts pitch via the lateral axis.

    III. Ailerons: Induce roll by moving in the same direction.

    IV. Flaps: Increase lift and thrust during takeoff.

    A. One statement is TRUE
    B. Two statements are TRUE
    C. Three statements are TRUE
    D. All statements are TRUE

7. In a four-stroke piston engine, which sequence is CORRECT?
   A. Intake → Compression → Power → Exhaust
   B. Compression → Intake → Exhaust → Power
   C. Power → Intake → Exhaust → Compression
   D. Compression → Intake → Power → Exhaust
8. During pre-flight inspection, ice is detected on the wing's leading edge. How does this primarily affect the wing's ability to generate lift?
   A. Increases form drag, reducing effective airspeed
   B. Disrupts airflow, lowering pressure differential
   C. Adds weight, shifting the center of gravity
   D. Alters angle of attack, causing inability to take off
9. Suppose an engine's spark plug malfunctioned which stroke of the piston power engine would be most impacted?
   A. Intake stroke
   B. Compression stroke
   C. Power stroke
   D. Exhaust stroke
10. A mechanic notice misplaced cams on the camshaft of a horizontally opposed engine. Which operational issue is most likely to result from this defect?
   A. Intake and exhaust valves timing errors
   B. Inconsistent fuel delivery to the combustion chamber
   C. Excessive vibration from unbalanced weight
   D. Overheating of the camshaft bearings
11. Which statements about lift production are TRUE?
   A. Bernoulli's Principle states that faster airflow reduces pressure
   B. A higher angle of attack always increases lift
   C. Wings generate lift solely due to their shape
   D. Two of the above statements are correct
12. Which statement(s) about glider flight are TRUE?

   I. Glide ratio is unaffected by wind conditions.

   II. A high-aspect-ratio wing minimizes induced drag.

   III. Adjusting angle of attack directly impacts lift-to-drag ratio.

   IV. Gliders cannot gain altitude during flight.

   A. One statement is TRUE
   B. Two statements are TRUE
   C. Three statements are TRUE
   D. All statements are TRUE
   E. None are TRUE

13. An aircraft experiences unwanted roll oscillations during turbulence. Which axis should the pilot prioritize adjusting?

A. Longitudinal axis
B. Lateral axis
C. Vertical axis

14. During a pre-flight check, a pilot notices the trim tab on the elevator is stuck. What immediate effect will this have on control inputs?

A. Reduced elevator responsiveness while pitching
B. Unwanted pitch during elevator movement
C. Unwanted pitch during aircraft approach
D. Increased physical effort to maintain pitch

15. A team designs a glider with a high-aspect-ratio wing to maximize glide ratio. Which two trade-offs are being impacted?

A. Increased induced drag and decreased parasite drag
B. Improved roll stability and worsen yaw stability
C. Decreased skin friction and increased form drag
D. Improved lift at low speeds and increased structural weight

16. An WWI aircraft with a rotary engine report difficulty executing tight turns. What design flaw inherent to rotary engines best explains this issue?

A. Stationary crankshaft
B. Spinning cylinders
C. Radial valve timings
D. Inefficient fuel delivery

17. During a rocket launch, engineers observe incomplete combustion in the engine's combustion chamber. What is the most likely cause of this issue?

A. Intake valve opening too early
B. Intake valve opening too late
C. Exhaust valve opening too early
D. Exhaust valve opening too late
E. Two of the above are correct

18. A pilot observes smoke in the exhaust and detects unburned fuel. Which part of the four-stroke engine is most likely malfunctioning?

A. Intake valve
B. Cylinder
C. Spark plug
D. Exhaust valve
E. None of the above

19. An aircraft designer is optimizing a commercial jet to reduce parasitic drag. Which combination of modifications would most effectively address both form drag and skin friction?

A. Installing winglets and polishing the fuselage to a mirror finish

B. Retracting landing gear during flight and applying riblet coatings to the wings

C. Streamlining antennae and washing the aircraft to remove contaminants

D. Using flush-mounted rivets and increasing the wingspan to distribute weight

20. All radio systems at an aerodrome fails during heavy rain. What communication method becomes most critical to maintain safe separation?

A. CATSA

B. ATC

C. NORDO

D. None of the above

21. An aircraft's oil includes viscosity modifiers. Under high operating temperatures, how do these modifiers impact oil viscosity?

A. Resisting excessive thinning

B. Resisting excessive thickening

C. Resisting excessive flowing

D. None of the above

22. A glider pilot adjusts the angle of attack to optimize glide ratio in stable atmospheric conditions. Which two aerodynamic effects will this adjustment most directly influence?

A. Induced drag increases and lift decreases

B. Parasite drag decreases and lift increase

C. Optimizes lift-to-drag ratio and wingtip vortices intensify

D. Form drag increases and skin friction decreases

23. An aircraft manufacturer decided to add extra parts to the empennage of an existing aircraft, what would happen to the aircraft's centre of gravity?

A. It would be shifted backwards

B. It would be shifted upwards

C. It would be shifted forwards

D. None of the above

24. During pre-flight checks, a mechanic identifies ice buildup on the wings. How does this condition primarily affect the aircraft's drag profile?

A. Increases induced drag by altering the wing's angle of attack

B. Decreases both form and interference drag through surface pressure

C. Decreases form drag by adding weight and destabilizing the aircraft

D. Increases parasite drag through increased skin friction and disrupted airflow

25. During a steep turn with heavy wind, a pilot notices increased resistance to forward motion. Which drag type is primarily responsible?

A. Induced drag
B. Form drag
C. Skin friction
D. Parasite drag

26. Regarding aircraft drag, which statement(s) are TRUE?

I. Induced drag is caused by wingtip vortices.
II. Parasite drag includes form drag and skin friction.
III. Streamlining reduces induced drag.
IV. Ice on wings increases parasite drag but reduces induced drag.

A. One statement is TRUE
B. Two statements are TRUE
C. Three statements are TRUE
D. Four statements are TRUE
E. None are TRUE

27. In a four-stroke piston engine, if the exhaust valve opens before intended during the power stroke, what is the most immediate consequence?

A. Unburned fuel enters the exhaust system
B. Compression efficiency worsens
C. The piston overheats due to internal pressure
D. The crankshaft rotation slows down

End of Exam.

1. D
2. C
3. D
4. B
5. A
6. B
7. A
8. B
9. C
10. A
11. A
12. B
13. A
14. D
15. D
16. B
17. C
18. C
19. C
20. C
21. A
22. C
23. A
24. D
25. B
26. B
27. A